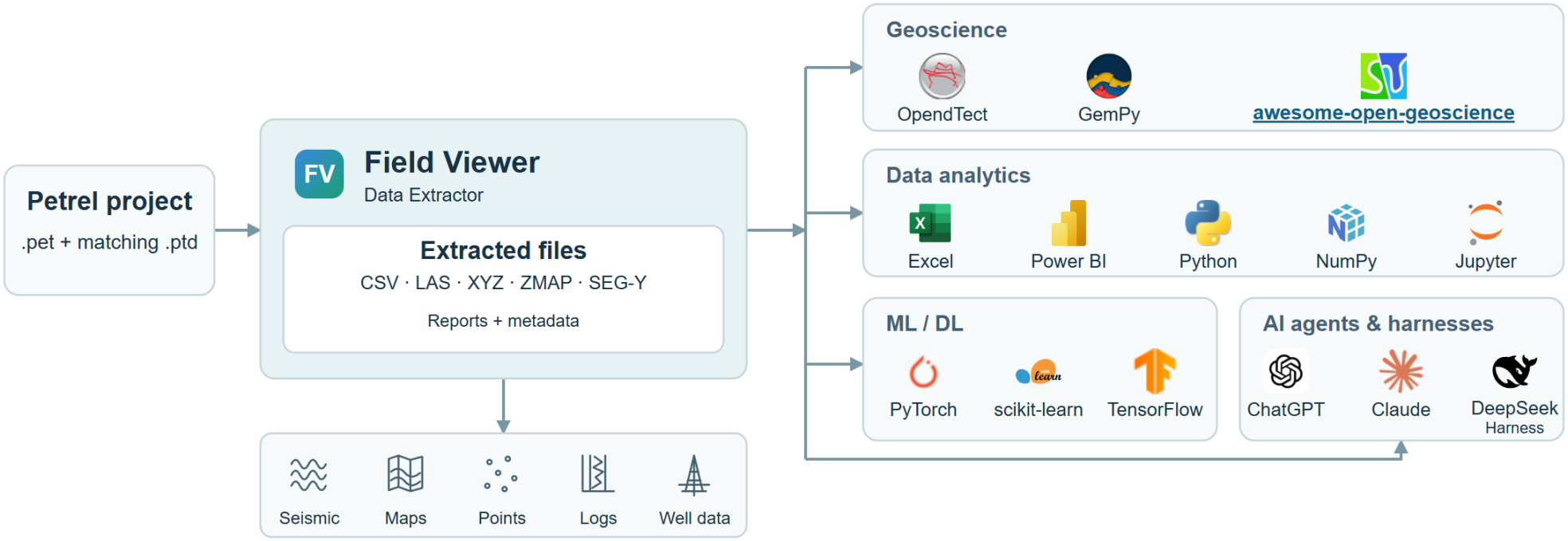
Start Here

Field Viewer Data Extractor

A small tool to help recover supported Petrel data for further work.

By Ahmed Saher Nouh / SaherLabs | Beta version (1.0.0)



GitHub repository | Download the Windows package | Release notes

The following pages walk through your first extraction.

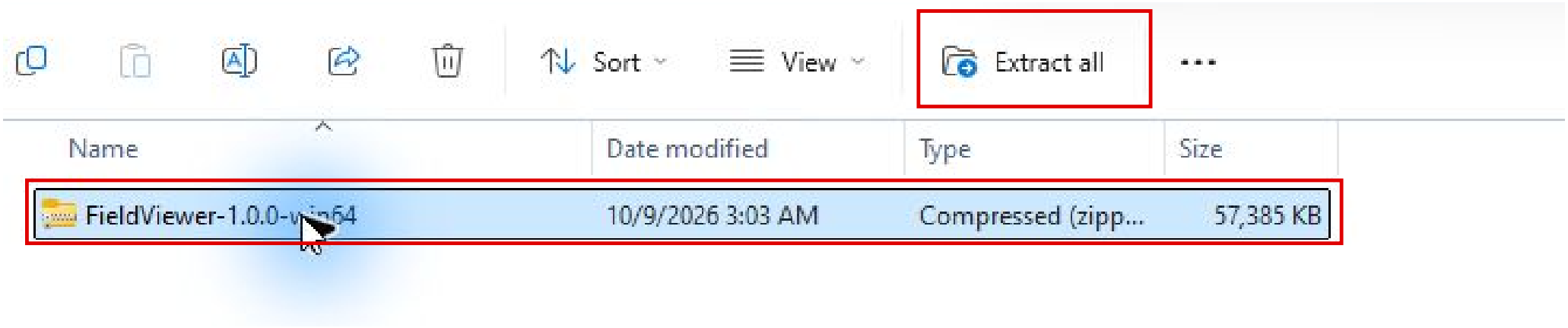
Start Here

GitHub: Project repository | Download the Windows package | Release notes

1. Extract the ZIP

You need Windows 64-bit, the complete application ZIP, a complete Petrel project and enough free storage for the outputs. Python and Petrel do not need to be installed separately. The screenshots use the SLB 2018 demo; use your own project path.

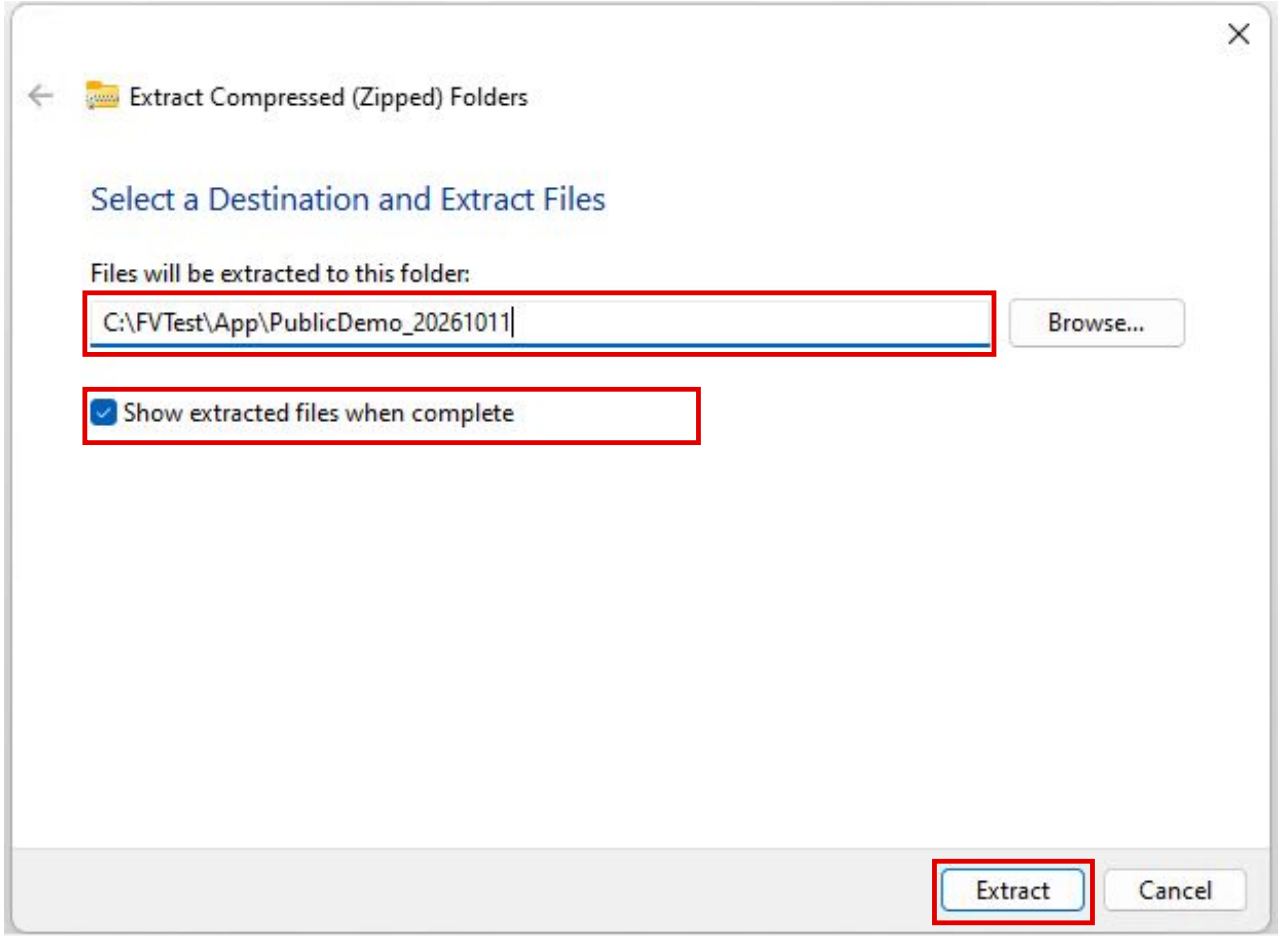
1. In File Explorer, select **the downloaded Field Viewer ZIP**. Click **Extract all**.
2. Extract the whole ZIP. Do not run the launcher from inside the compressed folder.



Select the ZIP, then click Extract all.

2. Choose the application folder

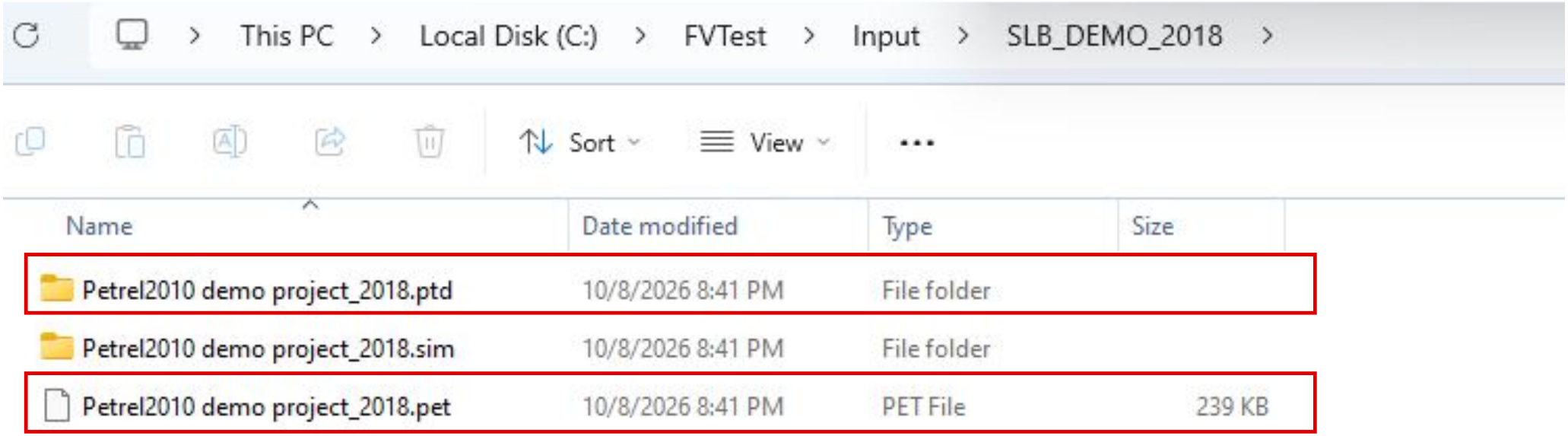
1. Enter a short, writable destination, for example **C:\FieldViewer**. Use a new folder if an earlier version is already present.
2. Leave **Show extracted files when complete** checked. Click **Extract**.



Destination, checkbox and Extract button.

3. Check your project files

1. Keep the **.pet** file beside its matching **.ptd** folder. Include every subfolder and seismic file.
2. Keep any other supplied project folders with the project. Use a copy and preserve the original files.

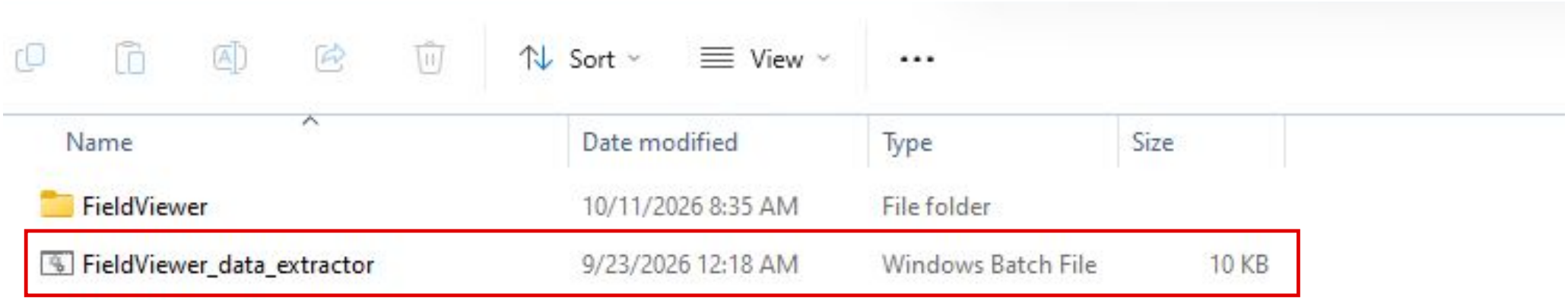


The .pet file and matching .ptd folder must stay together.

4. Open Field Viewer

1. In the extracted application folder, double-click **FieldViewer_data_extractor** (the Windows batch file). Windows may hide the **.bat** extension.

2. Keep the launcher beside the **FieldViewer** folder. Wait while the bundled runtime is prepared and checked.



Double-click the batch launcher.

5. Enter the full .pet path

1. At **Full path to the .pet project or .zgy file**, paste the complete **.pet** path, including the filename. Press **Enter**.

2. Use the location on your own computer. The example below is the demo path shown in the screenshot.

Demo example - replace this with your own full project path:

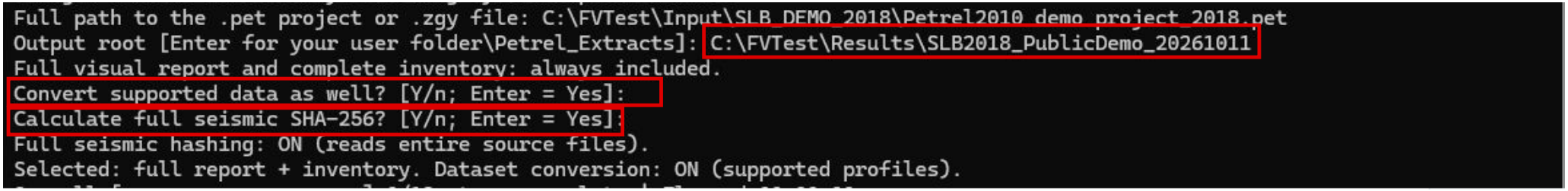
C:\FVTest\Input\SLB_DEMO_2018\Petrel2010 demo project_2018.pet



Paste the full project path at the prompt.

6. Choose the output and options

1. At **Output root**, enter a short folder outside the source project, then press **Enter**. Example: **C:\FVTest\Results\MyProject**.
2. At **Convert supported data as well?**, press **Enter** for Yes. At **Calculate full seismic SHA-256?**, press **Enter** for Yes.
3. The full report and inventory are always included. Keep the console open while extraction runs.



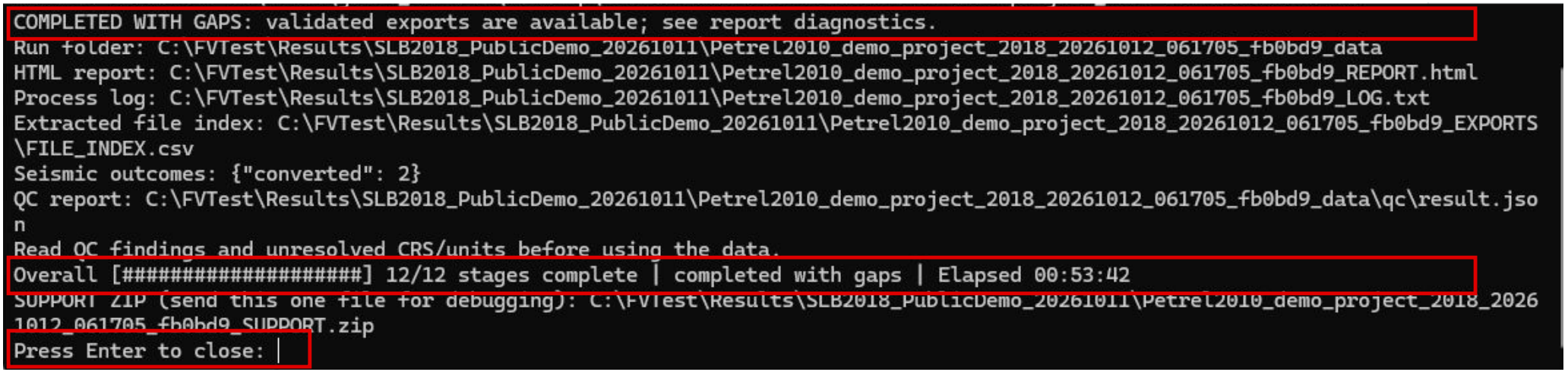
Enter an output folder and accept Yes at both option prompts.

7. Wait for the final result

1. Wait until the final status and output paths appear. Large projects can take a long time; allow the current run to finish.

2. **Completed with gaps** means outputs are available with stated limitations. Read the report before using them. Exit 10 by itself is not an installation failure.

3. After the final result and support ZIP are shown, press **Enter** to close the console. If the status is Failed, retain the logs and support ZIP and investigate before retrying.

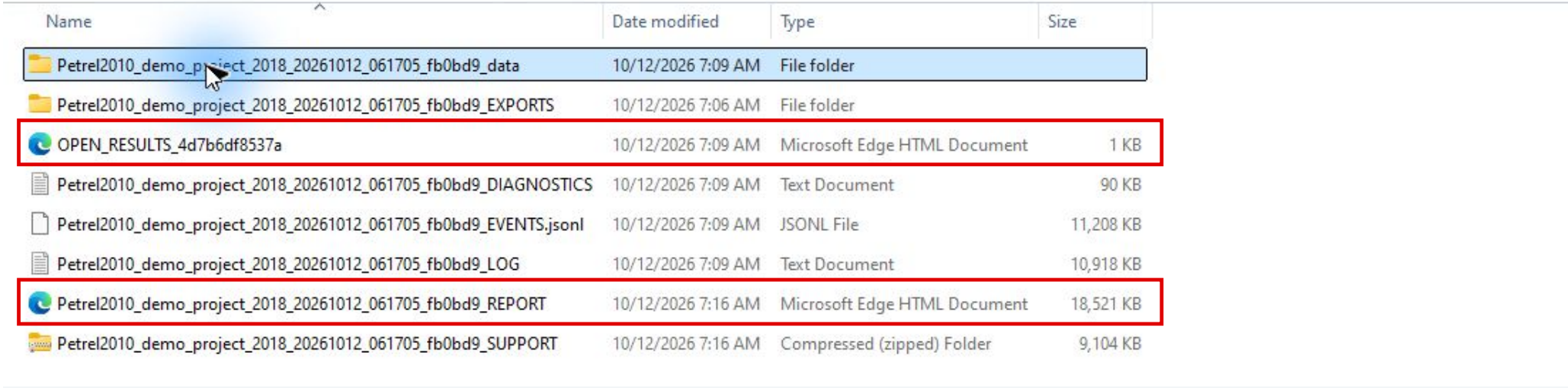


Read the final status before closing the console.

8. Open the HTML report

1. Open the output folder you chose. Double-click the file ending in **_REPORT.html**. Windows may display it without the extension.

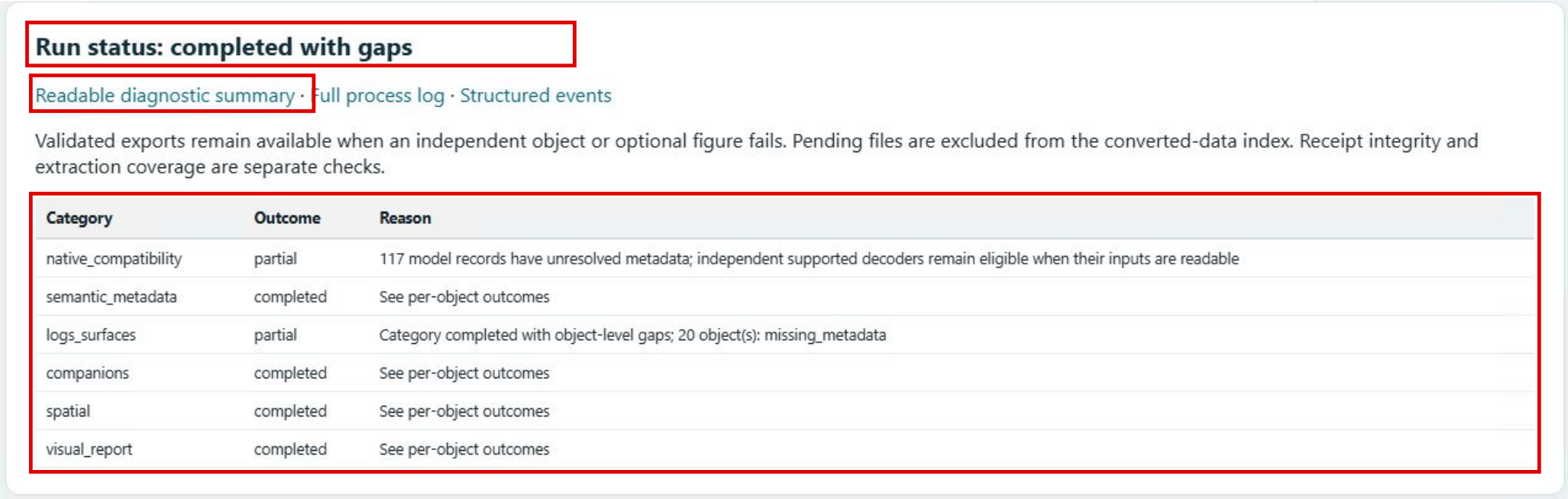
2. You can also open **OPEN_RESULTS**. Keep the report beside its **_EXPORTS** and **_data** folders so the links work.



Open the report file, or use OPEN_RESULTS.

9. Read the status and gaps

1. Read **Run status** and the category outcome table. Check what was recovered and what has missing or unsupported metadata.
2. Click **Readable diagnostic summary** for details when a category is partial. A completed run does not mean every object was decoded.



Run status: completed with gaps

Readable diagnostic summary · Full process log · Structured events

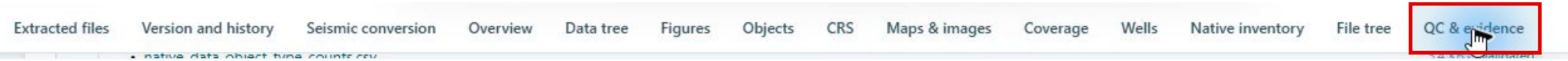
Validated exports remain available when an independent object or optional figure fails. Pending files are excluded from the converted-data index. Receipt integrity and extraction coverage are separate checks.

Category	Outcome	Reason
native_compatibility	partial	117 model records have unresolved metadata; independent supported decoders remain eligible when their inputs are readable
semantic_metadata	completed	See per-object outcomes
logs_surfaces	partial	Category completed with object-level gaps; 20 object(s): missing_metadata
companions	completed	See per-object outcomes
spatial	completed	See per-object outcomes
visual_report	completed	See per-object outcomes

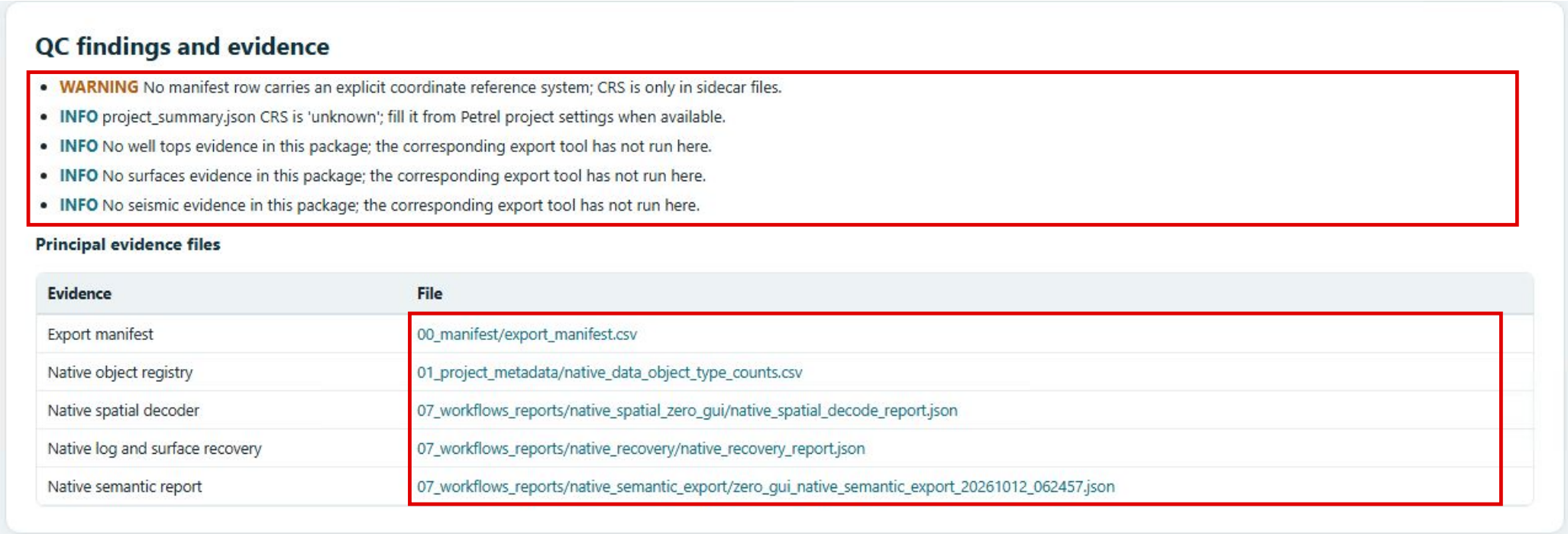
Read the final status, diagnostic link and category outcomes.

10. Check QC and metadata

1. Click **QC & evidence** in the report. Read the warnings and open the relevant evidence links.
2. Before importing a file, check its units, CRS and seismic axes in the matching **metadata.json**.



Click QC & evidence.



QC findings and evidence

- **WARNING** No manifest row carries an explicit coordinate reference system; CRS is only in sidecar files.
- **INFO** project_summary.json CRS is 'unknown'; fill it from Petrel project settings when available.
- **INFO** No well tops evidence in this package; the corresponding export tool has not run here.
- **INFO** No surfaces evidence in this package; the corresponding export tool has not run here.
- **INFO** No seismic evidence in this package; the corresponding export tool has not run here.

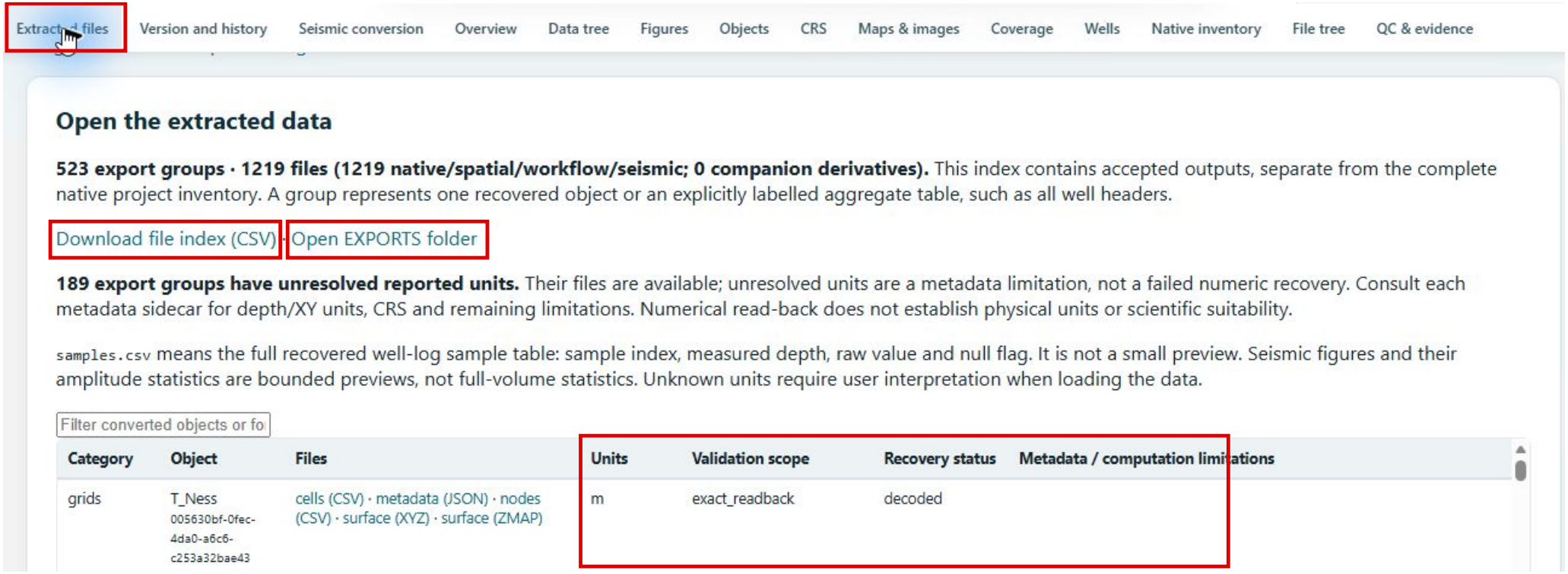
Principal evidence files

Evidence	File
Export manifest	00_manifest/export_manifest.csv
Native object registry	01_project_metadata/native_data_object_type_counts.csv
Native spatial decoder	07_workflows_reports/native_spatial_zero_gui/native_spatial_decode_report.json
Native log and surface recovery	07_workflows_reports/native_recovery/native_recovery_report.json
Native semantic report	07_workflows_reports/native_semantic_export/zero_gui_native_semantic_export_20261012_062457.json

Read the warnings and follow the relevant evidence links.

11. Open the extracted files

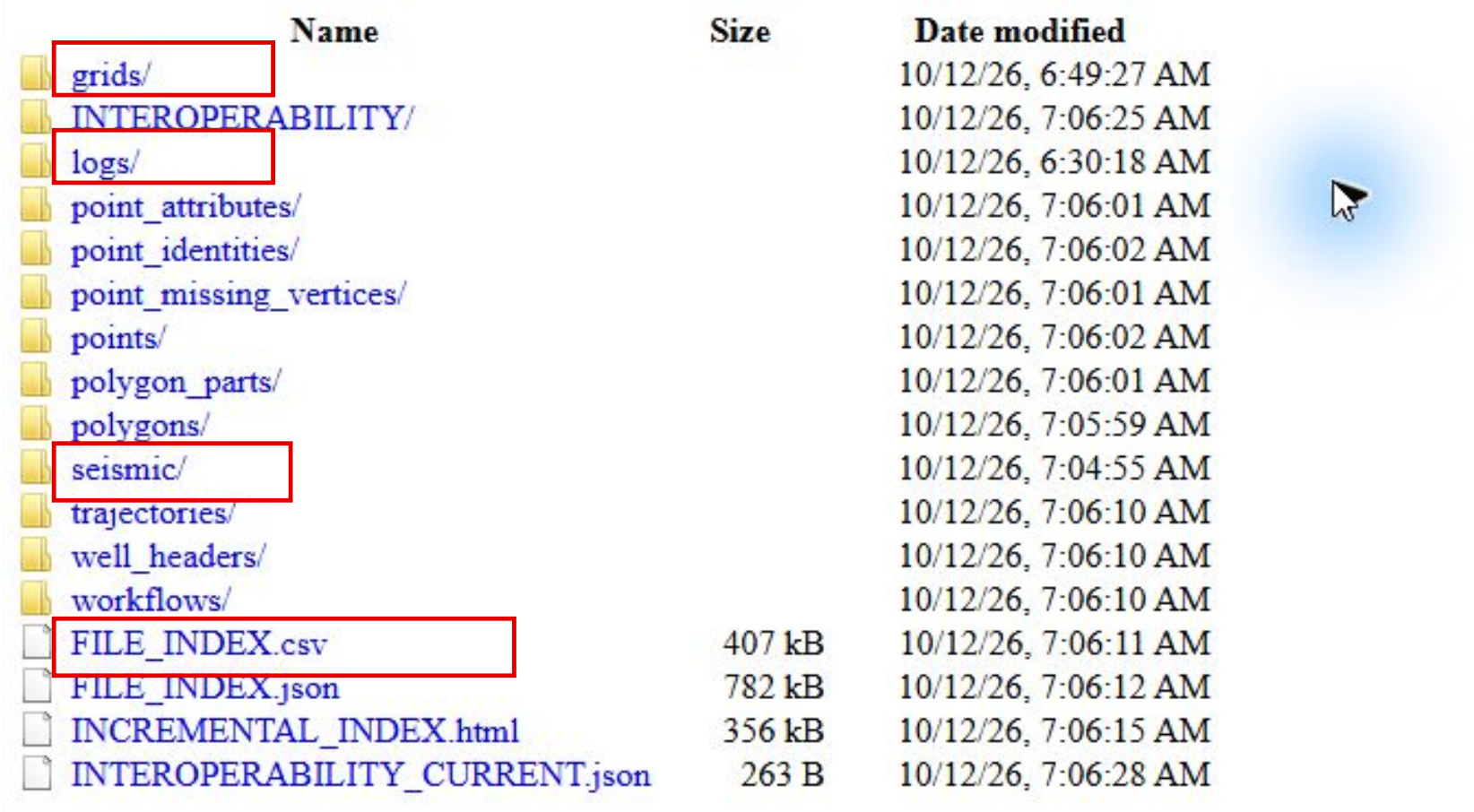
1. Click **Extracted files** in the report. Read the unit and recovery-status columns for the objects you need.
2. Click **Open EXPORTS folder** to browse the outputs. Use **Download file index (CSV)** to get the file list.
3. Keep data files with their metadata sidecars. Read unresolved-unit notices before using an export.



Use Extracted files, the CSV index and Open EXPORTS folder.

12. Find outputs by category

1. In **_EXPORTS**, open the category you need, such as **logs**, **grids** or **seismic**.
2. Open **FILE_INDEX.csv** for the full accepted file list. Filenames contain object identities to help match data to metadata.



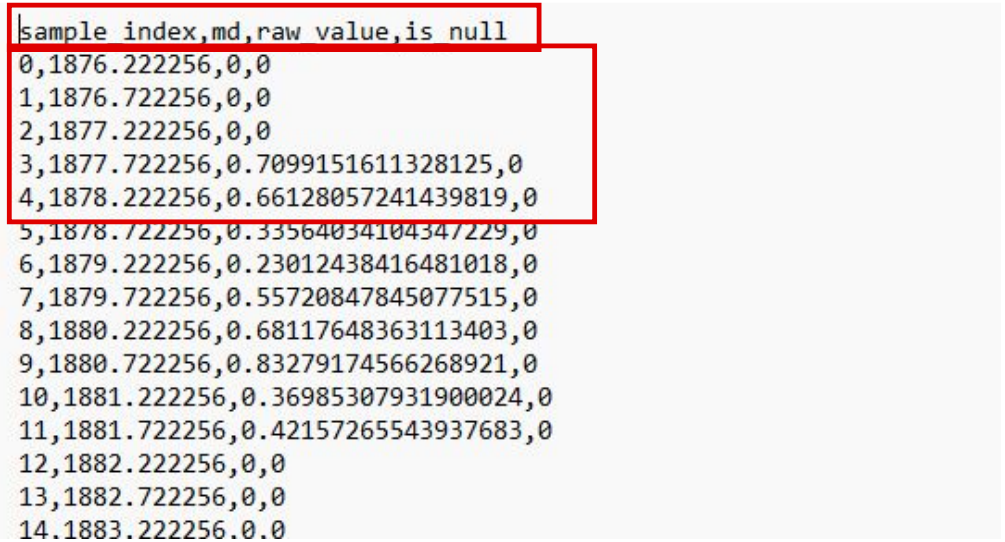
Name	Size	Date modified
grids/		10/12/26, 6:49:27 AM
INTEROPERABILITY/		10/12/26, 7:06:25 AM
logs/		10/12/26, 6:30:18 AM
point_attributes/		10/12/26, 7:06:01 AM
point_identities/		10/12/26, 7:06:02 AM
point_missing_vertices/		10/12/26, 7:06:01 AM
points/		10/12/26, 7:06:02 AM
polygon_parts/		10/12/26, 7:06:01 AM
polygons/		10/12/26, 7:05:59 AM
seismic/		10/12/26, 7:04:55 AM
trajectories/		10/12/26, 7:06:10 AM
well_headers/		10/12/26, 7:06:10 AM
workflows/		10/12/26, 7:06:10 AM
FILE_INDEX.csv	407 kB	10/12/26, 7:06:11 AM
FILE_INDEX.json	782 kB	10/12/26, 7:06:12 AM
INCREMENTAL_INDEX.html	356 kB	10/12/26, 7:06:15 AM
INTEROPERABILITY_CURRENT.json	263 B	10/12/26, 7:06:28 AM

Browse a category folder or open FILE_INDEX.csv.

13. Inspect data with its metadata

1. Open an exported **samples.csv** in a text editor or spreadsheet. Check the columns, raw values and null flags.

2. Open the matching **metadata.json**. Check **unit_status**, depth/axis units and **crs_status** for that object.



```
sample_index,md,raw_value,is_null
0,1876.222256,0,0
1,1876.722256,0,0
2,1877.222256,0,0
3,1877.722256,0.7099151611328125,0
4,1878.222256,0.66128057241439819,0
5,1878.722256,0.33564034104347229,0
6,1879.222256,0.23012438416481018,0
7,1879.722256,0.55720847845077515,0
8,1880.222256,0.68117648363113403,0
9,1880.722256,0.83279174566268921,0
10,1881.222256,0.36985307931900024,0
11,1881.722256,0.42157265543937683,0
12,1882.222256,0,0
13,1882.722256,0,0
14,1883.222256,0,0
```

Read the CSV column names and stored values.



```
"well": "C6",
"well_id": "bbaa2a07-0b4f-405e-bc2f-0d3a52a912b2",
"external_uwi": null,
"external_uwi_status": "unresolved; native well UUID is not an external UWI",
"template_id": "00000015-0000-0000-0000-000000000000",
"unit": "_",
"measurement": "Ratio",
"declared_template_unit": "_",
"unit_status": "validated_metric_profile",
"unit_resolution_gaps": [],
"depth_unit": "m",
"horizontal_unit": "m",
"project_units": {
  "xy": "m",
  "z": "m",
  "time": "ms",
  "system": "Metric"
},
"crs_status": "unresolved",
"source_sign_convention": "preserved; no time/depth sign reversal",
```

Read the matching unit fields and CRS status.

Red outline boxes show what to enter, click or read. Screenshots use the SLB 2018 demo.

GitHub: https://github.com/ahmedsahernouh/petrel-headless-extractor